

Introduction to Machine Learning and Well-Posed Learning Problems

(Your Name)

Why Machine Learning?

- Since computers were invented, we have wondered:
 - **Can we make computers learn from experience?**
- Goal: Program computers to **improve automatically with experience.**
- A successful theory of machine learning would:
 - Enable many new **applications** of computers.
 - Achieve new levels of **competence** and **customization**.
 - Help us understand **human learning abilities and disabilities.**

Motivating Examples

- **Medicine:** Learn from medical records which treatments are most effective for new diseases.
- **Smart homes:** Houses learn occupants' usage patterns and optimize energy costs.
- **Personal assistants:** Software agents learn evolving user interests and highlight relevant news stories.
- **Data mining:** Discover valuable knowledge from large commercial databases:
 - Equipment maintenance records
 - Loan applications
 - Financial transactions
 - Medical records, etc.

Current Status of Machine Learning

- We **do not yet** know how to make computers learn as well as humans.
- However:
 - Effective algorithms exist for many specific learning tasks.
 - A theoretical understanding of learning is **beginning to emerge**.
 - Many practical programs already exhibit **useful learning behavior**.
 - Significant **commercial applications** have begun to appear.
- For tasks like **speech recognition**, machine learning approaches:
 - Outperform all non-learning approaches tried so far.
- As computer science matures, it seems **inevitable** that ML will play a central role.

A Broad Definition of Learning

- We use a broad technical definition of learning:

Definition (Mitchell)

A computer program is said to learn from **experience** E with respect to some class of **tasks** T and **performance measure** P , if its performance at tasks in T , as measured by P , improves with experience E .

- Key ingredients:
 - Tasks T (what the system does)
 - Performance measure P (how we evaluate success)
 - Training experience E (data / interaction)

Well-Posed Learning Problems

- To have a **well-defined** learning problem, we must identify:
 - ① **Task** T : The class of tasks to be performed.
 - ② **Performance measure** P : How success on the task is measured.
 - ③ **Training experience** E : The source of experience or data.
- Without clearly specifying T , P , and E , it is unclear:
 - Whether the system is learning.
 - Whether performance is improving.

Example: Learning to Play Checkers

Checkers learning problem:

- **Task T :** Playing checkers.
- **Performance measure P :** Percent of games won against opponents.
- **Training experience E :** Playing practice games against itself.
- The program **learns** if:
 - Its win rate (as measured by P) improves as it gains more experience E .
- Typical setting:
 - Self-play generates large amounts of training data.
 - Strategies are updated based on game outcomes.

Example: Handwriting Recognition

Handwriting recognition learning problem:

- **Task T :** Recognizing and classifying handwritten words within images.
- **Performance measure P :** Percent of words correctly classified.
- **Training experience E :** A database of handwritten words with known (labeled) classifications.
- Goal: Learn a function mapping image patterns to word labels.
- Typical approach:
 - Use many labeled examples to train a classifier.
 - Evaluate on unseen words using P .

Example: Learning to Drive a Robot Car

Robot driving learning problem:

- **Task T :** Driving on public four-lane highways using vision sensors.
- **Performance measure P :** Average distance traveled before an error (as judged by a human overseer).
- **Training experience E :** Sequence of images and steering commands recorded while observing a human driver.
- The system learns a **driving policy** from human demonstrations.
- Applications: Autonomous vehicles, sensor-based control problems.

A Broad Class of Learning Programs

- The definition of learning is intentionally **broad**.
- It includes:
 - Intuitive learning tasks (e.g., recognizing images, playing games).
 - Simpler systems that improve through updates.
- Example:
 - A database system that allows users to update entries:
 - It improves its performance at answering queries about current data.
 - By the technical definition, it “learns” from experience.
- We do **not** focus on the everyday meaning of “learning”.
- Instead, we:
 - Precisely define a class of programs that improve with experience.
 - Study algorithms and theory for these programs.
 - Aim to understand the structure of learning problems and processes.

Some Successful Applications (1)

- **Speech recognition**

- Systems like SPHINX learn speaker-specific strategies.
- Techniques:
 - Neural networks.
 - Hidden Markov models (HMMs).
- Adapt to:
 - Different speakers and vocabularies.
 - Microphone characteristics.
 - Background noise.

- **Medical prediction**

- Programs that predict recovery rates of pneumonia patients.

Some Successful Applications (2)

- **Fraud detection**

- Detect fraudulent use of credit cards from transaction patterns.

- **Autonomous driving**

- Systems such as ALVINN:
 - Learned to drive unassisted at 70 mph for 90 miles on public highways.
 - Use visual input to generate steering commands.

- **Astronomical object classification**

- Decision tree learning used by NASA on the second Palomar Observatory Sky Survey.
- Automatically classifies celestial objects from ~ 3 terabytes of image data.

Some Successful Applications (3)

- **Game playing: Backgammon**

- TD-GAMMON learned by playing over one million practice games against itself.
- Achieved performance competitive with the human world champion.
- Demonstrates:
 - Learning effective strategies in huge search spaces.

- **General pattern**

- Machine learning provides powerful tools for:
 - Large search spaces.
 - Complex patterns in data.
 - Dynamic environments.

- Theoretical results characterize fundamental relationships among:
 - Number of training examples observed.
 - Number (or complexity) of hypotheses under consideration.
 - Expected error of the learned hypothesis.
- This leads to:
 - Understanding how much data is needed.
 - Trade-offs between model complexity and generalization.
 - Formal learning frameworks (e.g., PAC learning, VC dimension).

Links to Human and Animal Learning

- Initial models of **human and animal learning** are being developed.
- Researchers study:
 - How human/animal learning compares to machine learning algorithms.
 - How insights from cognitive science and neurobiology can inspire new algorithms.
- Examples:
 - Cognitive architectures.
 - Neural-network models motivated by brain studies.
- Overall:
 - Applications, algorithms, theory, and biological studies all show **rapid progress** over the last decade.

Disciplines Influencing Machine Learning (1)

Artificial Intelligence

- Learning symbolic representations of concepts.
- Viewing learning as a **search** problem.
- Using prior knowledge together with training data to guide learning.

Bayesian Methods

- Bayes' theorem as the basis for computing probabilities of hypotheses.
- Naive Bayes classifier.
- Algorithms for estimating unobserved variables (probabilistic inference).

Computational Complexity Theory

- Theoretical bounds on the difficulty of learning tasks.
- Effort measured by:
 - Computational cost.
 - Number of examples.
 - Number of mistakes, etc.

Disciplines Influencing Machine Learning (2)

Control Theory

- Learning to control processes to optimize objectives.
- Learning to predict the next state of the controlled process.

Information Theory

- Measures of entropy and information content.
- Minimum Description Length (MDL) principle in learning.
- Relationship between optimal codes and optimal training sequences.

Philosophy

- Occam's razor: Prefer the simplest hypothesis consistent with the data.
- Justification for generalizing beyond observed data.

Psychology and Neurobiology

- Power law of practice: human response time improves with practice according to a power law.
- Neurobiological studies motivate artificial neural network models.

Statistics

- Characterizing errors when estimating hypothesis accuracy from limited data:
 - Bias and variance.
- Tools:
 - Confidence intervals.
 - Statistical significance tests.

Summary

- Machine learning studies **programs that improve with experience**.
- A well-posed learning problem specifies:
 - Task T , performance measure P , and experience E .
- Many successful applications:
 - Speech recognition, medical prediction, fraud detection, autonomous driving, astronomy, game playing, and data mining.
- Theory provides relationships between:
 - Data size, hypothesis complexity, and expected error.
- Machine learning is inherently **multidisciplinary**, drawing on:
 - AI, Bayesian methods, complexity, control, information theory, philosophy, psychology, neurobiology, and statistics.
- Goal of the field:
 - Understand and design algorithms that can **learn effectively from experience**.